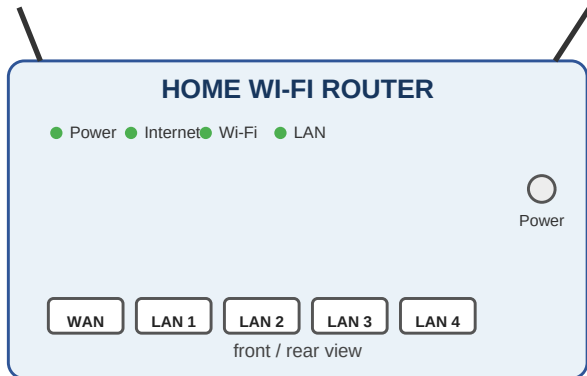


Router & Network Switch

Assignment Answers

1. Labelled diagram of a typical home Wi-Fi router and network switch

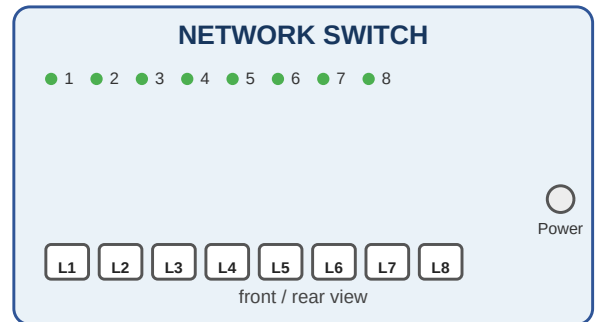


Router labels:

WAN = ISP/Internet connection

LAN = wired local-network ports

LEDs = power, Internet, Wi-Fi and LAN status



Switch labels:

LAN ports = connect PCs/devices

Port LEDs = link/activity status

Power = DC power input

Illustration: port positions vary by manufacturer and model.

Important: A router usually has a WAN/Internet port for the ISP connection and several LAN ports for the local network. A switch mainly provides multiple Ethernet ports for devices on the same LAN. The status LEDs show conditions such as power, link/activity, Internet and Wi-Fi status.

2. Three key differences between routers and switches

Difference	Router	Switch
Main function	Connects different networks, such as the home LAN and the Internet, with routes and forwards packets between them.	Connects devices within the same local network and forwards packets between them.
Address used for forwarding	Uses IP addresses and routing information to decide where packets should go.	Uses MAC addresses and a MAC address table to send Ethernet frames to the correct port.
Type of network connected	Normally connects separate networks or subnets, such as a corporate WAN or the Internet.	Normally connects devices in a LAN, such as PCs, printers, and servers.

In simple terms: A router decides how traffic moves *between networks*; a switch decides which local Ethernet port should receive traffic *within a network*.

3. Identifying ports and LED indicators

The following are virtual reference illustrations of common router and switch layouts. Exact port locations differ between models, but the names and purposes are typical.

Item	Router — what to identify	Switch — what to identify
Power port	Usually a round DC barrel jack on the rear, connected to the power supply. Capable of supplying electrical power to the router.	Usually a standard AC power outlet on the rear, connected to the power supply. Capable of supplying electrical power to the switch.
Ethernet ports	WAN/Internet port connects to the ISP modem/ONT. LAN ports connect to devices (e.g., PCs, laptops, tablets, and other networked devices).	Multiple Ethernet ports (e.g., 8, 16, 24, or 48) for connecting devices to the switch.
LED indicators	Common LEDs show Power, Internet/WAN, Wi-Fi, and LAN activity. A flashing LED port indicates activity.	Common LEDs show Power, Port Link/Speed, and Port Activity. A flashing LED port indicates activity.

Annotation guide: On a physical device, first locate the power socket, then identify the WAN/Internet port (router) and at least two LAN Ethernet ports. Look beside or above the ports for LEDs labelled Power, Internet, Wi-Fi, Link, Act or Speed. On switches, port LEDs are commonly placed directly above or below each Ethernet port.

Example image source for real devices: Wikimedia Commons has photographs of routers and Ethernet switches that can be used for visual reference: [Router images](#) and [Ethernet switch images](#).

4. Small gaming café network scenario

Scenario: A small gaming café needs reliable Internet access for IPL streaming and multiplayer gaming on 8 PCs. The ISP provides an Ethernet handoff or a modem/ONT. I would connect the ISP line to the **router's WAN/Internet port first**. The router then creates the café's local network, provides IP addressing (DHCP), and routes traffic between the LAN and the Internet.

Connecting the 8 gaming PCs: I would connect one of the router's LAN ports to an Ethernet port on an **8-port or larger Gigabit Ethernet switch**. Then I would connect each of the 8 gaming PCs to a separate switch port using Ethernet cables. If the router has limited LAN ports, the switch provides the additional wired ports needed for all PCs.

Connection	Purpose
ISP line → Router WAN	Brings Internet service into the café and lets the router route traffic between the ISP and the LAN
Router LAN → Switch	Extends the wired LAN and provides many Ethernet ports for the gaming PCs.
Switch → Gaming PCs 1–8	Gives each PC a direct wired Ethernet connection for stable, low-latency local connectivity.
Router → Wi-Fi / other devices	Can provide Wi-Fi for staff phones, customer devices or a separate access point if needed.

Why this choice? The router belongs at the network edge because it connects the café's private LAN to the ISP. The switch is then used for the 8 gaming PCs because switches are designed to connect many wired devices on the same LAN. For gaming, wired Ethernet also avoids the variable wireless conditions that can occur with Wi-Fi. A Gigabit switch is appropriate for modern gaming PCs and local network traffic.

Simple layout: ISP → **Router (WAN)** → **Router LAN** → **Switch** → PC1, PC2, PC3, PC4, PC5, PC6, PC7, PC8.